TRIPATHI STUDY ZONE
JEE & NEET
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CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 2

(With Solutions)

Duration : 3 Hours **Max. Mark : 720**

Instructions

- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- The question paper contains three parts of Physics, Chemistry and Biology respectively.
- Part A contains 45 Questions from Physics section. Part B contains 45 Questions from Chemistry section. Part C contains 90 Questions from Biology.

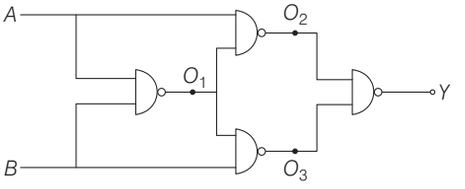
PART A PHYSICS

1. Force required to move a train of 1500 quintals up an incline of 1 in 75 (a slope that has a rise of 1 unit for every 75 unit of displacement) with an acceleration of 4 ms^{-2} is
(where, force of friction = 0.2 newton per quintal)
(a) 650000 N (b) 690000 N
(c) 619900 N (d) 625000 N

2. In a sonometer experiment, density of the material of wire used is 7.5×10^3 kg m^{-3}. If the stress in the wire is 3×10^8 Nm^{-2}, the speed of transverse wave on wire, is
(a) 225 ms^{-1} (b) 250 ms^{-1}
(c) 200 ms^{-1} (d) 100 ms^{-1}

3. A radioactive sample with a half life of 1 yr has the label 'activity = 5μ curie on 1-9-2018'. Then, what would be its activity two years earlier?
(a) 0.5 μ curie (b) 10 μ curie
(c) 20 μ curie (d) 50 μ curie

4. The output of the given logic circuit is



(a) $\overline{A}\overline{B} + BA$ (b) $(A + B)\overline{A}$
(c) $\overline{A} \cdot B + \overline{B} \cdot A$ (d) None of these

5. Relation between wavelength of photon and electron of same energy is
(a) $\lambda_{ph} > \lambda_e$ (b) $\lambda_{ph} < \lambda_e$
(c) $\lambda_{ph} = \lambda_e$ (d) λ_e / λ_{ph} = constant

6. A steam boat goes across a lake and comes back (*A*) on a quite day, when the water is still and (*B*) on a rough day, when there is uniform air current, so as to help the journey onward and to impede the journey back.

If the speed of the launch on both days was same, in which case it will complete the journey in lesser time?

(a) Case (*A*) (b) Case (*B*)
(c) Same in both (d) Nothing can be predicted

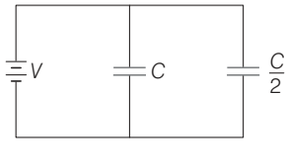
7. Two coherent light beams of intensities I and $4I$ are superposed. The maximum and minimum possible intensities in the resulting beam are

(a) $5I$ and I (b) $5I$ and $3I$
(c) $9I$ and I (d) $9I$ and $3I$

8. Indian space agency (ISRO) thinking about making space elevator on earth by using steel cables. If an untapered space elevator cable would need a material capable of sustaining a length of 5000 km of its own weight at sea level to reach a geostationary altitude of 35800 km without yielding. So, for this idea, the length of the steel cable which will break under its own weight when suspended vertically is (take, breaking strength of steel = $1.2\times 10^{9}\ \mathrm{Nm^{-2}}$, density of steel cable = $8\times 10^{3}\ \mathrm{kg/m^{3}}$ and gravitational acceleration, $g = 10\ \mathrm{m/s^{2}}$)

(a) 20,000 m (b) 50 km (c) 15000 km (d) 15 km

9. Two condensers, one of capacity C and the other of capacity $\frac{C}{2}$, are connected to V volt battery, as shown in the figure.



The work done in charging fully both the condensers is

(a) CV^2 (b) $\frac{1}{4}CV^2$ (c) $\frac{3}{4}CV^2$ (d) $\frac{1}{2}CV^2$

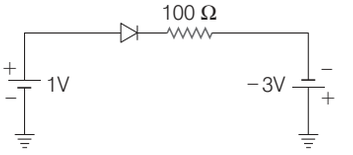
10. A carnot engine is a device

(a) which convert mechanical energy into heat energy
(b) which convert heat energy into mechanical energy
(c) absorbs heat from a sink at a lower temperature and rejects to the source at high temperature
(d) None of the above

11. A black body of mass 50 g and surface area $11.46\ \mathrm{cm^{2}}$ is at an initial temperature of 800 K. It is allowed to cool inside an evacuated enclosure kept at constant room temperature 300 K. The rate of cooling is 0.05°C/s. The specific heat of body is (take, Stefan's constant, $\sigma = 5.73\times 10^{-8}\ \mathrm{Jm^{-2}\,K^{-4}}$)

(a) $11.5\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$ (b) $15.5\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$
(c) $9.5\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$ (d) $10.5\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$

12. What is the current through an ideal p-n junction diode shown below?



(a) Zero (b) 10 mA (c) 20 mA (d) 40 mA

13. A star subtends an angle ϕ at the pole of a concave mirror of focal length f. Then, the radius of the image of the star formed by the mirror is

(a) $\frac{2}{f\phi}$ (b) $\frac{f\phi}{2}$
(c) $2\phi f$ (d) ∞

14. The frequency f of a stretched string depends upon the tension force F, length l of the string and the mass per unit length μ of the string. The formula for frequency is (k = a constant)

(a) $f = \frac{k}{l}\sqrt{\frac{\mu}{F}}$ (b) $f = \frac{k}{l}\sqrt{\frac{F}{\mu}}$
(c) $f = kl\sqrt{\frac{\mu}{F}}$ (d) $f = kl\sqrt{\frac{F}{\mu}}$

15. The internal energy of an ideal diatomic gas decreases by the same amount as the work done by the system

(i) the process must be adiabatic
(ii) the temperature must decrease
(iii) the process must be isobaric
(iv) the process must be isothermal

Which option satisfies the statements?

(a) (i), (iv) (b) (ii), (iii)
(c) (i), (ii) (d) All statements are correct

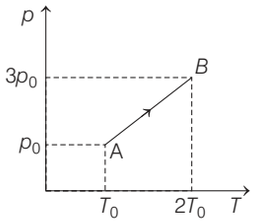
16. A particle of mass 10g is describing SHM along a straight line with a period of 2s and amplitude of 10 cm. The difference in kinetic energy, when it is 2 cm and 5 cm from its equilibrium position is

(a) $100\pi^2$erg (b) $105\pi^2$erg
(c) $480\pi^2$erg (d) $375\pi^2$erg

17. A particle of mass 10 g is kept on the surface of an uniform sphere of mass 100 kg and radius 10 cm. Then, the work to be done to take the particle far away from the sphere is
(take, $G = 6.67\times 10^{-11}\ \mathrm{N\text{-}m^{2}\,kg^{-2}}$)

(a) 3.33×10^{-10} J
(b) 6.67×10^{-10} J
(c) 13.34×10^{-10} J
(d) 6.67×10^{-9} J

18. Pressure *versus* temperature graph of an ideal gas is shown in the figure. Density of the gas at point A is ρ_0. Density at point B will be



(a) $\frac{3}{4}\rho_0$ (b) $\frac{3}{2}\rho_0$ (c) $\frac{4}{3}\rho_0$ (d) $2\rho_0$

19. Two soap bubbles of diameter d_A and d_B combine under isothermal conditions in space outside international space station at low earth orbit. Now, find the new radius of bubble, if low earth orbit at a distance of 37000 km from earth.

(a) $\frac{d_A + d_B}{2}$ (b) $\frac{d_A d_B}{d_A + d_B}$

(c) $\sqrt{d_A d_B}$ (d) $\sqrt{d_A^2 + d_B^2}$

20. Two bodies of masses 10 kg and 2 kg are moving with velocity $(-2\hat{\mathbf{i}} + 7\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$ and $(10\hat{\mathbf{i}} - 35\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$, respectively. The velocity of the centre of mass will be

(a) $2\hat{\mathbf{k}}\ \text{ms}^{-1}$ (b) $2\hat{\mathbf{j}}\ \text{ms}^{-1}$ (c) $3\hat{\mathbf{k}}\ \text{ms}^{-1}$ (d) $3\hat{\mathbf{j}}\ \text{ms}^{-1}$

21. A ractangular coil having 20 turns carries a current of 15 A. The coil is oriented with respect to uniform magnetic field of 0.2 T, such that its area vector is $-0.05\,\hat{\mathbf{i}}\ \text{m}^2$. The potential energy of coil in the given orientation is

(a) 0 (b) 7.5 J (c) 4 J (d) 3 J

22. In an adiabatic compression of a monoatomic gas and its volume reduces to one-third of initial volume, then its final pressure would be, if gas has initial pressure p

(a) $3^{1/2}\ p$ (b) $3^{3/2}\ p$

(c) 6.24 p (d) $3^{3/5}\ p$

23. An air bubble inside a cubical block of glass of side 7 cm appears to be at 3.5 cm from one face and 1.5 cm from the other face opposite to the first when viewed normally. Then, the real depth of the bubble from the opposite face is [take, refractive index of glass, $\mu = 1.5$]

(a) 3 cm (b) 2.5 cm (c) 2.1 cm (d) 2 cm

24. A point mass oscillates along the X-axis according to the relation $x = x_0 \cos(\omega t - \pi/4)$. If acceleration of the particle is written as $a = a_0 \cos(\omega t + \delta)$, then

(a) $a_0 = x_0\omega^2,\ \delta = -\pi/4$ (b) $a_0 = x_0\omega^2,\ \delta = \pi/4$

(c) $a_0 = x_0\omega^2,\ \delta = -3\pi/4$ (d) $a_0 = x_0\omega^2,\ \delta = 3\pi/4.$

25. Match the quantities of Column I with their values in Column II and choose the correct option from the codes given below

	Column I		Column II
A.	Planck constant	1.	$\frac{h}{p}$
B.	Stopping potential	2.	$E_0 - \text{KE}_{\text{max}}$
C.	Work function	3.	$\frac{\text{KE}_{\text{max}}}{e}$
D.	de-Broglie wavelength	4.	$\frac{E}{\nu}$

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	2	4	(b)	3	2	1	4
(c)	4	3	1	2	(d)	4	3	2	1

26. The threshold wavelength of photon for a metal surface is 36 Å . If a light of wavelength 2240 Å is incident on metal surface, then velocity of ejected electron will be
(given, $h = 6.63 \times 10^{-34}$, $c = 3 \times 10^8$ m/s, $m_e = 9.1 \times 10^{-31}$kg)

(a) 18.4×10^6 m/s (b) 8.6×10^5 m/s

(c) 15×10^6 m/s (d) 12.5×10^5 m/s

27. The reading of a spring balance corresponds to 100 N, while situated at the north pole and a body is kept on it. The weight recorded on the same scale, if it is shifted to the equator is
(take, acceleration due to gravity at poles $g_p = 10\ \text{ms}^{-2}$ and radius of earth = 6.4×10^3 m)

(a) 99.66 N (b) 110 N (c) 97.66 N (d) 106 N

28. Two similar coils of radius r and number of turns N are lying concentrically with their planes at right angles to each other. The current flowing in them are $\sqrt{6}\ I$ and $\sqrt{3}\ I$ respectively, then magnetic field induction at centre of the coils is

(a) $\frac{3\mu_0 NI}{4r}$ (b) $\frac{\mu_0 NI}{4r}$ (c) $\frac{3\mu_0 NI}{2r}$ (d) $\frac{\mu_0 NI}{2r}$

29. At normal temperature and pressure, the speed of sound in air is 330 ms^{-1}. The speed of sound in hydrogen at 273ºC and 3 atm pressure is (take, air as 16 times heavier than hydrogen)

(a) 1850 ms^{-1} (b) 1860 ms^{-1}

(c) 1866.48 ms^{-1} (d) 1806.48 ms^{-1}

30. A voltmeter has a resistance of 90 Ω is connected across a cell of emf 10 V and internal resistance 10 Ω . The voltmeter reads the voltage of

(a) 12 V (b) 6 V

(c) 3 V (d) 9 V

31. If Young's double slit experiment is performed in water instead of air, then
(a) no fringes would be seen
(b) fringe width would decrease
(c) fringe width would increase
(d) fringe width would remain unchanged

32. A current loop of radius r has a magnetic field induction B_1, at a point on its axis at a distance r from its centre and B_2 at its centre, then $B_1 : B_2$ is
(a) $1 : 2\sqrt{2}$ (b) $2\sqrt{2} : \sqrt{3}$
(c) $1 : \sqrt{2}$ (d) $\sqrt{3} : 2\sqrt{2}$

33. The voltage and current through a circuit element are $V = 100 \sin(314t + 45^\circ)$ V,
$i = 10 \sin(314t + 315^\circ)$A. Then, circuit element is
(a) capacitor (b) inductor
(c) resistor (d) None of these

34. An aluminium sphere of radius 10 cm is heated from 0° to 100°C. Its volume changes by (where, coefficient of linear expansion for aluminium = $23 \times 10^6/^\circ$ C)
(a) 28.9 cc (b) 2.89 cc
(c) 9.28 cc (d) 49.8 cc

35. A projectile P is projected at an angle of 30° with the horizontal. If a second projectile Q is projected such that their speeds are in ratio 3 : 1 for same maximum height, then the angle of projection of Q is
(a) $\sin^{-1}\left(\frac{3}{1}\right)$ (b) $\sin^{-1}\left(\frac{1}{3}\right)$
(c) $\sin^{-1}\left(\frac{3}{2}\right)$ (d) $\sin^{-1}\left(\frac{2}{3}\right)$

36. In a region, electric field and magnetic field both are acting in the same direction. If a proton is projected along the direction of the fields with a certain velocity, then
(a) it will turn towards left (b) it will turn towards right
(c) its velocity will increase (d) its velocity will decrease

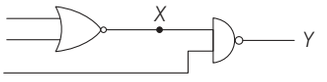
37. The earth has a mass of 6×10^{24} kg and a radius of 6.4×10^6 m. The amount of work done in joules, if its rotation were to be increased, so that the duration of the day becomes 20 h instead of 24 h. (Moment of inertia of earth = $\frac{2}{5} MR^2$)
(a) -9.360×10^{25} J (b) 9.360×10^{30} J
(c) 9.358×10^{28} J (d) -9.358×10^{30} J

38. A current loop of area $0.05\ m^2$ and carrying a current of 4 A is held perpendicular to a magnetic field induction of 0.5 T, then torque acting on the loop is
(a) 0.01 N-m (b) 0.1 N-m
c) 1 N-m (d) 0 N-m

39. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall, for its two different positions. The area of the source of light is
(a) $\frac{1}{A_1} + \frac{1}{A_2}$ (b) $\frac{A_1 - A_2}{2}$ (c) $\sqrt{A_1 A_2}$ (d) $\frac{A_1 + A_2}{2}$

40. The mass of jupiter is 315 times that of earth and the diameter of jupiter is 11.5 times that of earth. If value of g on earth is $9.8\ ms^{-2}$, then its value on jupiter is
(a) $23.52\ ms^{-2}$ (b) $23.90\ ms^{-2}$
(c) $23.60\ ms^{-2}$ (d) $23.34\ ms^{-2}$

41. The output Y, when all the three inputs are first high and then low, will respectively be



(a) 1, 1 (b) 1, 0 (c) 0, 1 (d) 0, 0

42. Which of the following statement(s) is (are) correct regarding electric lines of forces?
(i) The electric lines of forces are parallel for non-uniform electric field.
(ii) Closer are the electric lines of forces, stonger is the field, and further apart are lines of force, weaker is electric field.
(iii) Electric lines of forces never cross each other otherwise, there will be two dircetions of electric field at a point of intersection which is not possible.
(a) (i) only (b) (i) and (ii) only
(c) (ii) and (iii) only (d) all are correct

43. In gamma ray, emission from a nucleus
(a) Both the neutron number and the proton number change
(b) there is no change in the proton number and the neutron number
(c) only the neutron number changes
(d) only the proton number changes

44. If potential difference across a capacitor is changed from 10 V to 20 V, then work done is W. What will be the work done, if potential difference is changed from 30 V to 40 V?
(a) $10W$ (b) W (c) $\frac{W}{2}$ (d) $2W$

45. A research satellite of mass 200 kg circles the earth in an orbit of average radius $\frac{3R}{2}$, where R is the radius of earth. Assuming the gravitational pull on a mass of 1 kg on the earth's surface to be 10 N, the pull on the satellite will be
(a) 880 N (b) 889 N
(c) 885 N (d) 892 N

PART B CHEMISTRY

46. If travelling at same speeds, which of the following matter waves have the shortest wavelength?
(a) Electron (b) α-particle
(c) Neutron (d) Proton

47. The value of n in
$MnO_4^- + 8H^+ + ne^- \longrightarrow Mn^{2+} + 4H_2O$ is
(a) 5 (b) 4 (c) 2 (d) 3

48. In a flask at a certain temperature there are 2 g H_2 and 8 g O_2. The mole fraction of O_2 in the given mixture is
(a) 8/5 (b) 0.2 (c) 0.25 (d) 1.0

49. Which of the following statements is in accordance with the Arrhenius equation?
(a) Rate of a reaction has no effect with increase in temperature
(b) Rate of a reaction increases with decrease in activation energy
(c) Rate constant decreases exponentially with increase in temperature
(d) Rate of reaction decreases with decrease in activation energy

50. Helium atom is two times heavier than hydrogen molecule. At 298 K, the average kinetic energy of helium atom is
(a) two times that of hydrogen molecule
(b) same as that of hydrogen molecule
(c) four times that of hydrogen molecule
(d) half that of hydrogen molecule

51. The work to be done on 2 moles of perfect gas at 27°C, if it is compressed reversibly and isothermally from a pressure of 1.01×10^5 Nm^{-2} to $5.05 \times 10^6 Nm^2$ is
(a) 2.03×10^4 J (b) 1.95×10^2 J
(c) 1.95×10^4 J (d) 1.95×10^8 J

52. pH of 0.5 M aqueous solution of HF ($K_a = 2 \times 10^{-4}$) is
(a) 2 (b) 4 (c) 6 (d) 10

53. An electrochemical cell consists of two half-cell reactions

$$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$$
$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$$

The mass of copper (in grams) dissolved on passing 0.5 A current for 1 hour is (Given : Atomic mass of Cu is 63.6; $F = 96500$ C mol^{-1})
(a) 0.88 g (b) 1.18 g
(c) 0.29 g (d) 0.56 g

54. Solubility of AgCl in water, 0.01 M $CaCl_2$, 0.01 M NaCl and 0.05 M $AgNO_3$ are S_1, S_2, S_3 and S_4 respectively, then
(a) $S_1 < S_2 < S_3 < S_4$ (b) $S_1 > S_3 > S_2 > S_4$
(c) $S_1 > S_2 = S_3 > S_4$ (d) $S_1 > S_3 > S_4 > S_2$

55. Among following reactions, which one will produce dihydrogen?
(a) $H_2S_4O_8 + H_2O$ (b) $BaO + HCl$
(c) $Mg + H_2O$ (d) $Na_2O_2 + 2HCl$

56. Which of the following alkaline earth metals do not impart any colour to the flame?
(a) Ca, Sr (b) Mg, Ca
(c) Be, Mg (d) Sr, Ba

57. The solution which maintains its pH constant even upon addition of small amounts of acid or base, is called buffer solution. Which can act as a buffer?
(a) $NH_4Cl + NH_4OH$
(b) $CH_3COOH + CH_3COONa$
(c) 40 mL of 0.1 M NaCN + 20 mL of 0.1 M HCN
(d) All of the above

58. The metallic luster exhibited by sodium is explained by the
(a) existence of body centred cubic lattice
(b) excitation of free electrons
(c) diffusion of sodium ions
(d) oscillation of loose electrons

59. Which one has maximum oxidation member of the underlined atoms in the following compounds?
(a) $\underline{Mn}O_4^{2-}$ (b) $\underline{Cr}O_5$
(c) $\underline{Cr}O_2Cl_2$ (d) Equal in all the options

60. Perkin's condensation is an example of chemical properties of aromatic aldehydes only. In this reaction aromatic aldehydes condense with an acid anhydride to yield α,β-unsaturated acids. Intermediate involved in Perkin reaction is
(a) carbocation (b) carbanion
(c) free radical (d) carbene

61. The per cent loss in mass after heating a pure sample of 245 g potassium chlorate (mol. mass =122.5) will be
(a) 12.25 (b) 24.50
(c) 39.18 (d) 49.0

62. Anhy. $AlCl_3$ produces fumes in the air. This is due to
(a) hydrolysis (b) reduction
(c) oxidation (d) dimerisation

63. Quartz is extensively used as a pizeoelectric material as it contains

(a) Sn (b) Ti
(c) Si (d) Pb

64. Basic strength of $CH \equiv \bar{C}$(I); $CH_2 = \bar{C}H$ (II) and $CH_3\bar{C}H_2$ (III) will be in order

(a) I <II<III (b) II <III> I
(c) III <II<I (d) III <I<II

65. The temperature of the slag zone in the metallurgy of iron using blast furnace is

(a) 1200-1500°C (b) 1500-1600°C
(c) 400-700°C (d) 800-1000°C

66. The one of the product obtained on reaction of ethyl magnesium bromide with methanol is

(a) ethane (b) methane
(c) propane (d) methoxyethane

67. For ZnS, the coordination number of cation and anion are respectively

(a) 4,4 (b) 4,6
(c) 6,6 (d) 4,8

68. Which of the following statements is/are incorrect regarding corrosion?

(a) Impurities present in iron enhance rusting by settling a number of miniature cells
(b) It is an electrochemical phenomenon
(c) The technique of protecting a metal from corrosion by connecting it to a second metal is known as anodic protection
(d) It is an oxidation deterioration of a metal

69. Which of the following represents the correct increasing order of acidic behaviour of the molecules given below ethyne, benzene and hexane?

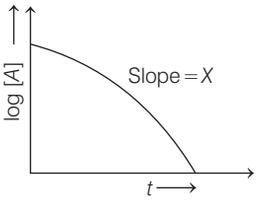
(a) Benzene < Hexane < Ethyne
(b) Hexane < Benzene < Ethyne
(c) Hexane < Benzene > Ethyne
(d) Ethyne < Hexane < Benzene

70. Which of the following curve show Freundlich adsorption isotherm?

(a) $\log \frac{x}{m}$ vs $\log p$ (b) $\log \frac{x}{m}$ vs $\log p$

(c) $\log \frac{x}{m}$ vs $\log p$ (d) $\log \frac{x}{m}$ vs $\log p$

71. A graph of log [*A*] *vs t* for first order reaction is given below.



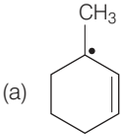
X is equal to

(a) $\frac{0.693}{k}$ (b) $\frac{k}{2.303}$ (c) $-\frac{k}{2.303}$ (d) $\log [A]_0$

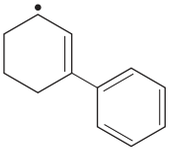
72. For the reaction $A \rightarrow B$, $\frac{-d[A]}{dt} = \frac{2d[B]}{dt}$ then the rate law is

(a) $-\frac{d[B]}{dt} = k[B]^2$ (b) $\frac{d[B]}{dt} = k[A]$
(c) $\frac{d[A]}{dt} = k[A]^2$ (d) $-\frac{d[A]}{dt} = k[A]^2$

73. Which of the following is the most stable free radical?

(a) 

(b) 

(c) 

(d) 

74. $CoSO_4Cl \cdot 5NH_3$ exists in two isomeric forms '*A*' and '*B*'. Isomer '*A*' reacts with $AgNO_3$ to give white precipitate, but does not react with $BaCl_2$. Isomer '*B*' gives white precipitate with $BaCl_2$ but does not react with $AgNO_3$. The IUPAC name of '*A*' and '*B*' respectively are

(a) [*A*] is pentaamminesulphatocobalt (III) chloride
[*B*] is pentaamminechloridocobalt (III) sulphate
(b) [*A*] is tetraamminedisulphatocobalt (III) chloride
[*B*] is pentaamminechloridecobalt (III) sulphate
(c) [*A*] is triamminedisulphatocobalt (III) chloride
[*B*] is pentaamminechloridocobalt (III) sulphate
(d) [*A*] is pentaamminesulphatocobalt (III) chloride
[*B*] is tetraamminedichloridocobalt (III) sulphate

75. Solids are attracted by magnetic field due to the presence of atoms, ions or molecules with unpaired electron, called paramagnetic. Which among the following is paramagnetic?

(a) Cl_2O_7 (b) Cl_2O
(c) ClO_2 (d) Cl_2O_5

76. Which of the following statement(s) regarding actinides is/are correct?

(a) Pu^{4+} disproportionates to Pu^{3+} and PuO_2^{2+} in strongly acidic medium

(b) Maximum oxidation state of Np is +7.

(c) UO_2^{2+} is stable

(d) All of the above statements are correct

77. Count the total number of S—O bonds, which are having equal length in bisulphate ion?

(a) 3 (b) 4 (c) 5 (d) 6

78. The synthesis of alkyl fluorides is best accomplished by

(a) Swart's reaction (b) free radical fluorination

(c) Finkelstein reaction (d) Sandmeyer reaction

79. In dichromate ion ($Cr_2O_7^{2-}$), each Cr is linked to

(a) two O- atoms (b) three O- atoms

(c) four O- atoms (d) five O- atoms

80. The coordination entity formed when excess of KCN is added to an aqueous solution of $CuSO_4$ is

(a) $[Cu(CN)_2]^+$ (b) $[Cu(CN)_6]^{4-}$

(c) $[Cu(CN)_4]^{2-}$ (d) $[Cu(CN)_4]^{3+}$

81. Which of the following solvent is used in the synthesis of chlorofluorocarbon?

(a) Methylene chloride

(b) Carbon tetrachloride

(c) Chloroform

(d) Iodoform

82. A solution containing 2.665 g of $CrCl_3 \cdot 6H_2O$ is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of $AgNO_3$ to give 2.87 g of AgCl. The structure of compound is

(a) $[CrCl \cdot (H_2O)_5]Cl_2 \cdot H_2O$

(b) $[CrCl_2 \cdot (H_2O)_4]\ Cl \cdot H_2O$

(c) $[CrCl \cdot (H_2O)_5]Cl \cdot H_2O$

(d) $[CrCl_2 \cdot (H_2O)_5]Cl_2 \cdot H_2O$

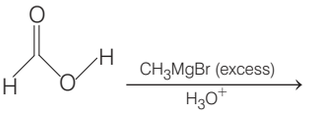
83. Match the Column I with Column II and choose the correct option using the codes below.

	Column I		Column II
A.	O_2^-	1.	Paramagnetic
B	CN^-	2.	Diamagnetic
C.	CO	3.	Diamagnetic
D.	$\overset{+}{N}O$	4.	Diamagnetic

Codes

	A	B	C	D		A	B	C	D
(a)	2	3	4	1	(b)	3	4	2	1
(c)	1	2	3	4	(d)	4	3	2	1

84. The major product of the following reaction is



(a) Propanol (b) Propanal

(c) Acetaldehyde (d) None of these

85. Which one of the following pairs of pyrimidine bases are present in DNA?

(a) Cytosine and uracil

(b) Cytosine and guanine

(c) Cytosine and adenine

(d) Cytosine and thymine

86. Phenol is heated with pthalic anhydride in presence of conc. H_2SO_4. The product gives pink colour with alkali. The product is

(a) phenolpthalein (b) bakelite

(c) salicylic acid (d) fluorescein

87. For the heterogenous reaction, $A(s) + 2B(g) \rightleftharpoons 3D(s) + 2Y(g)$, the relationship between equilibrium constants, K_p and K_C is

(a) $K_p = K_C [RT]^2$ (b) $K_p = K_C [RT]^0$

(c) $K_p = K_C [RT]^{-2}$ (d) $K_p = K_C [RT]^{-1}$

88. Among the aldehydes and ketones, "the size of alkyl group is inversely proportional to their reactivity". Cannizzaro reaction is an example of disproportionation reaction, i.e. self oxidation-reduction. Which of the following will undergo Cannizzaro reaction?

(a) Benzaldehyde (b) Acetaldehyde

(c) Acetone (d) Pentanone

89. Those polymers which get decomposed by the process of biodegradation are known as biodegradable polymer. Among the following, the biodegradable polymer is

(a) cellulose (b) polyethene

(c) PVC (d) nylon-6

90. CH_3CONH_2 and $CH_3CH_2NH_2$ are distinguished by reacting with

(a) *aq*. NaOH and heat (b) *aq*. HCl and heat

(c) Br_2 water (d) acidic $KMnO_4$

PART C BIOLOGY

91. Leaves and branches falling from trees often damage smaller plants beneath them. It is an example of
(a) amensalism (b) commensalism
(c) mutualism (d) competition

92. In bryophytes, the food conducting cells are known as
(a) hydroids (b) leptoids (c) rhizoids (d) bulbils

93. Which among the following statements is/are correct regarding eutrophication?
(a) It is natural ageing of water body
(b) It occurs due to nutrient enrichment of water body particularly with nitrogen and phosphorus
(c) Both (a) and (b)
(d) None of the above

94. Which of the following plants can grow in N_2 deficient soil?
(a) *Nymphaea* (b) *Pisum sativum*
(c) *Stellaria* (d) *Nepenthes*

95. The bacterium commonly called 'Weizman organism' helps in the fermentation of sugar to a mixture of organic solvents for the production of biofuel blend stock is
(a) *Clostridium acetobutylicum* (b) *Pseudomonas nitrificans*
(c) *Acetobacter aceti* (d) *Thiobacillus denitrificans*

96. The biopatents are awarded for achievements in
I. strains of microorganisms.
II. DNA sequences.
III. proteins encoded by DNA sequences.
IV. cell lines.
Choose the correct option.
(a) I and II (b) I and III
(c) III and IV (d) I, II, III and IV

97. Match the era given in Column I with the organisms that appeared in these periods in Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Devonian	1.	Fishes
B.	Ordovician	2.	Man
C	Pennsylvanian	3.	Reptiles
D.	Holocene	4.	Amphibians

Codes

	A	B	C	D
(a)	4	1	3	2
(c)	2	4	1	3

	A	B	C	D
(b)	1	3	4	2
(d)	3	2	4	1

98. Which among the following statements is incorrect regarding dicotyledons?
(a) Possess tap root system
(b) Leaves have reticulate venation
(c) Stem is branched
(d) Secondary growth is absent

99. Thermocycler is used in
(a) micropropagation (b) hybridisation
(c) PCR (d) fermentation

100. Identify the incorrectly matched pair?
(a) J-shaped curve – Logistic growth
(b) S-shaped curve – Represented under limited resource conditions
(c) Both (a) and (b)
(d) None of the above

101. During eukaryotic transcription, what is the role or function of an enhancer?
(a) Increases *m*RNA stability
(b) Elongation of transcript
(c) Transcription initiation
(d) Stimulates the rate of transcription

102. Which of the following is a freshwater sponge that harbours green algae in it?
(a) *Euspongia*
(b) *Euplectella*
(c) *Cliona*
(d) *Spongilla*

103. Which factor(s) among the following is/are responsible for decrease in absorption of water by the roots at low temperature?
(a) increased viscosity of water
(b) decreased rate of root elongation
(c) decreased metabolic activity of root cells
(d) All of the above

104. Choose the option that correctly explains the meaning of the terms related to ecological succession.

(a)	Sere	–	Sequence of communities that successively change in a given area
(b)	Seral stage	–	Changes during ecological succession that leads to a community
(c)	Climax community	–	Individual transitional communities
(d)	Pioneer community	–	Gradual and predactable change in a community

105. With respect to the mechanism by which the plant tissues sense gravity (gravisensing), which of the following statements is correct?

(a) It leads to redistribution of auxin in stems and roots, leading to differential growth
(b) It may be involving pressure-sensitive channels at the top and bottom of cells
(c) Starch grains inside of cells which fall to the lower side of the cell (statoliths) may be involved
(d) All are correct

106. Which class of cyclin-CDK complex is present at the G_2 checkpoint and is required for the cell cycle to enter the mitotic phase from G_2-phase?

(a) CdK_1/Cyclin B (b) CdK_4/Cyclin D
(c) CdK_6/Cyclin D (d) CdK_4/Cyclin B

107. Match the following columns and choose the correct option from the codes given below.

	Column I		Column II
A.	PCT	1.	Urine concentration
B.	Vasa recta	2.	Reabsorption
C.	Glomerulus	3.	Ultrafiltration
D.	DCT	4.	Selective secretion

Codes

	A	B	C	D		A	B	C	D
(a)	2	4	3	1	(b)	4	1	3	2
(c)	2	1	3	4	(d)	1	2	3	4

108. The two kingdom system of classification was discarded because of the

(a) presence of characters of both plants and animals in *Euglena*
(b) difference of bacteria and cyanobacteria from other organisms
(c) non-photosynthetic nature of fungi
(d) All of the above

109. The host cells are made competent to take up DNA by treating them with

(a) potassium (b) strontium
(c) magnesium (d) calcium

110. Which of the following techniques is/are highly effective for birth control?

I. Vasectomy II. Tubectomy
III. Periodic abstinence IV. Coitus interruptus

Choose the correct option.

(a) I and II (b) III and IV
(c) II and III (d) Only IV

111. Which of the following exhibits myxotrophic nutrition?

(a) *Euglena* (b) Cyanobacteria
(c) *Amoeba* (d) Sponges

112. The left phallomere, a part of the external genitalia of male *Periplaneta* consists of which of the following sets of structures?

(a) Titillator, phallic gland
(b) Phallic gland, mushroom gland
(c) Vasa deferentia, an ejaculatory duct
(d) Titillator, pseudopenis

113. In protein structure, the strongest bonds are

(a) ionic bonds (b) covalent bonds
(c) hydrogen bonds (d) van der Waals interactions

114. The maximum transportation of oxygen in the blood occurs

(a) in dissolved form (b) as bicarbonate
(c) as oxyhaemoglobin (d) Both (a) and (b)

115. Which one among the following statements correctly describes a tracheid?

(a) Lignin is deposited in the cellulose wall to form annular or spiral thickenings
(b) Dead, single cells with hard and lignified walls
(c) Both (a) and (b)
(d) None of the above

116. The tuberculated rhizoids are characterised by

(a) narrow rhizoids
(b) lack of cytoplasm at maturity
(c) peg-like ingrowth
(d) All of the above

117. Which of the following fungi is used for the production of antibiotic penicillin?

(a) *S. griesus*
(b) *P. notatum*
(c) *P. chrysogenum*
(d) Both (b) and (c)

118. CAM (Crassulacean Acid Metabolism) plants have so adapted, that allows them to

(a) function better in humid conditions
(b) fix CO_2 during the day
(c) save enzyme RuBisCO from denaturation
(d) survive in arid conditions

119. Match the following columns and choose the correct option from the codes given below.

	Column I		Column II
A.	*Enterobius vermicularis*	1.	Whipworm
B.	*Wuchereria bancrofti*	2.	Eyeworm
C.	*Trichuris trichiura*	3.	Filarial worm
D.	*Loa loa*	4.	Pinworm

Codes

	A	B	C	D		A	B	C	D
(a)	4	3	1	2	(b)	3	1	2	4
(c)	2	3	4	1	(d)	4	2	1	3

120. Deficiency of which of the following is the cause for albinism?
(a) Acetyl cholinesterase (b) Carbonic anhydrase
(c) Tyrosinase (d) Glutamine

121. Hybrid vigour (or heterosis) is a result of
(a) homozygosity of the gametes
(b) blending of male and female characteristics
(c) increased heterozygosity in the offspring
(d) retention of ancestral genes

122. In collenchyma, the corners of cells are deposited with
(a) cellulose (b) pectic substances
(c) Both (a) and (b) (d) None of these

123. Which of the following is a major inorganic chemical component of Murashige and Skoog's nutrient medium?
(a) Potassium nitrate (b) Sodium EDTA
(c) Calcium sulphate (d) Cobalt chloride

124. The book written by Carolus Linnaeus, which lists every species of plant known at that time is
(a) *Systema Naturae* (b) The origin of life
(c) Green Data Book (d) *Species Plantarum*

125. Which among the following is not an example of biopiracy?
(a) Basmati rice (*Oryza sativa*)
(b) Black pepper (*Piper nigrum*)
(c) Indian mustard (*Brassica campestris*)
(d) *Bacillus anthracis*

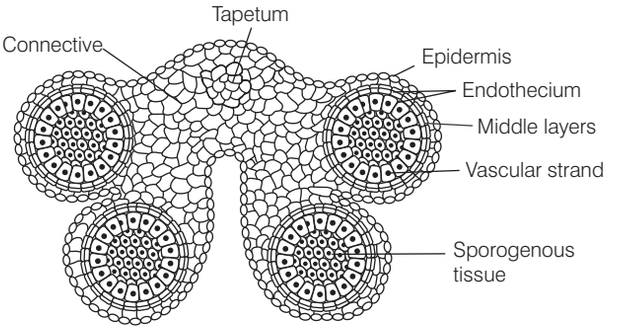
126. A human gene was amplified, cloned and finally transformed into *E.coli*. But, gene expression turned out to be poor because of
(a) codon bias
(b) presence of termination signals
(c) presence of introns
(d) All of the above

127. Genes whose expression (product) is constantly required for cellular activity are known as
(a) housekeeping genes
(b) smart genes
(c) regulatory genes
(d) structural genes

128. A series of genes, transcription of which is regulated by DNA binding proteins
(a) promoter (b) operator
(c) operon (d) terminator

129. Diuretic drugs that promote secretion of excess urine from the body, are commonly used in the treatment of
(a) high blood pressure (b) renal failure
(c) renal calculi (d) glomerulonephritis

130. The figure given below is showing the structure of a microsporangium. Two of its labellings are wrong. Choose the option which correctly indicates them.



Structure of microsporangium

(a) Tapetum and vascular strand
(b) Endothecium and connective
(c) Middle layers and sporogenous cells
(d) Tapetum and endothecium

131. Identify the correct statement(s) regarding bulliform cells.
(a) Bulliform cells are found in monocot leaves
(b) They help in rolling of leaves in dry weather
(c) They are also known as motor cells
(d) All of the above

132. Ephemerals are short-lived xerophytes, which lack the capacity to tolerate drought. They survive in the critical dry period in the form of
(a) barks and leaves (b) leaves and thorns
(c) fruits and seeds (d) thorns and seeds

133. The psychological disorder characterised by unreasonable thoughts and fear, repetitive behaviours and panic attack is called
(a) clinical depression
(b) Obsessive Compulsive Disorder (OCD)
(c) bipolar disorder
(d) attention deficit disorder

134. Match the following 'man' given in Column I with their cranial capacity given in Column II and choose the correct option from the codes given below.

Column I	Column II
A. Handy man	1. 900 cc
B. Peking man	2. 1360 cc
C. Java ape man	3. 700 cc
D. Cro-Magnon man	4. 1075 cc

Codes

	A	B	C	D
(a)	1	2	3	4
(c)	3	4	1	2

	A	B	C	D
(b)	2	3	4	1
(d)	4	1	2	3

135. Which among the following statement(s) is/are correct regarding dioxin?
(a) It is a colourless organic compound
(b) It is the most toxic herbicide
(c) It was used in Vietnam War as a defoliating agent
(d) All of the above

136. An annelid which shows bioluminescence is
(a) *Polynoe* (b) *Lumbricus*
(c) *Chaetopterus* (d) *Nereis*

137. Two species of rose grow in the same garden but do not interbreed because one blooms in winters and one blooms in summers. This is best explained by
(a) temporal isolation (b) mechanical isolation
(c) geographical isolation (d) gametic isolation

138. The vital capacity includes
(a) inspiratory reserve volume + tidal volume
(b) tidal volume + inspiratory reserve volume + expiratory reserve volume
(c) expiratory reserve volume + tidal volume
(d) expiratory reserve volume + inspiratory reserve volume

139. In which type of placenta, the villi are present in the form of transverse bands?
(a) Discoidal (b) Diffused
(c) Cotyledonary (d) Zonary

140. Which of the following statements is/are correct?
(a) HBOT can be used for wound healing
(b) Oxygen is a critical factor in wound healing
(c) Both (a) and (b)
(d) None of the above

141. The oxides of sulphur are an important pollutant because they cause
(a) drying up of margin of leaf and portion of lamina in between vascular bundles
(b) harm to eyes, throat and lungs
(c) Both (a) and (b)
(d) None of the above

142. In certain body parts like finger tips, nose, pinnae, etc., their is direct connection between the arteriole and venule, which is known as,
(a) arterio-venous anastomosis
(b) interelastic membrane
(c) diapedesis
(d) pleuripotential haemopoietic stem cells

143. The leader region of the *trp* operon codes for RNA that may function as a
(a) inducer (b) repressor
(c) attenuator (d) apoprotein

144. The basic structure of liver is composed of which tissue?
(a) Mucous tissue (b) Reticular tissue
(c) Areolar tissue (d) Lymphoid tissue

145. Heavy Mero-Myosin (HMM) contains
(a) globular head
(b) tail
(c) globular head with a short arm
(d) short arm and a tail

146. Mycoplasma
(a) is a prokaryote and lacks cell wall
(b) exhibits pleomorphism
(c) does not show crystallisation
(d) All of the above

147. 'Gomphoses' is an example of
(a) non-movable joints (b) imperfect movable joints
(c) perfect movable joints (d) sutures

148. Mycorrhiza is the symbiotic relationship between ..... and roots of plants.
(a) algae (b) cyanobacteria
(c) fungi (d) bacteria

149. Which one of the following statements is false with respect to the condition phenylketonuria?
(a) It is the result of pleiotropy
(b) It is an autosomal recessive trait
(c) It is a metabolic error
(d) It is a case of aneuploidy

150. When a gene has two or more alleles that are common in a population, the phenomenon is known as
(a) polyploidy (b) polygenic inheritance
(c) polymorphism (d) complementary genes

151. The sympathetic nervous system regulates all of these except
(a) gall bladder (b) liver
(c) adrenal glands (d) urinary bladder

152. Match the Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Dicot stem	1.	Hypodermis is sclerenchymatous
B.	Monocot stem	2.	Pericycle gives rise to lateral roots only
C.	Monocot root	3.	Pericycle gives rise to lateral roots and secondary meristem
D.	Dicot root	4.	Hypodermis is collenchymatous

Codes

	A	B	C	D
(a)	1	2	3	4
(b)	1	4	3	2
(c)	1	4	2	3
(d)	4	1	2	3

153. Name the class of antiviral proteins produced by T-lymphocytes after viral stimulation.
(a) Alpha interferon (b) Gamma interferon
(c) Beta interferon (d) None of these

154. The injury to vagus nerve is unlikely to affect which of the following?

I. Movement of eyelids.
II. Tongue movement.
III. Cardiac movement.
IV. Peristaltic movement.

Choose the correct option.

(a) I, III and IV (b) I and II
(c) III and IV (d) Only II

155. Which among the following statement(s) is/are correct regarding cymose inflorescence?

(a) Development of flowers is basipetal
(b) The growth is definite
(c) Opening of flowers is centrifugal
(d) All of the above

156. A boy whose metabolic rate is abnormally high and has protruding eyeball is suffering from which of the following diseases?

(a) Goitre (b) Grave's disease
(c) Hashimoto's disease (d) Addison's disease

157. For the development of root and stem, a plant requires

(a) calcium (b) copper
(c) magnesium (d) sodium

158. Match the pancreatic hormones given in Column I with the cells that are responsible for their secretion given in Column II.

	Column I		Column II
A.	Insulin	1.	δ-cells
B.	Glucagon	2.	α-cells
C.	Somatostatin	3.	γ-cells
D.	Pancreatic polypeptide	4.	β-cells
		5.	F-cells

Choose the correct option from the codes given below.

Codes

	A	B	C	D		A	B	C	D
(a)	4	2	1	5	(b)	1	5	4	2
(c)	4	2	5	1	(d)	1	3	2	4

159. Biomagnification occurs because

(a) pollutants like DDT, mercury are non-biodegradable
(b) pollutants like DDT cannot be metabolised by organisms
(c) pollutants like DDT cannot be excreted out by organisms consuming them
(d) All of the above

160. A person is suffering from deficiency of a certain hormone due to which his I.Q. level is equivalent to a 5 years old boy. Which hormone is deficient in this man?

(a) Aldosterone (b) GH
(c) ACTH (d) Thyroxine

161. Match Column I with Column II and choose the correct option from the codes given below.

	Column I		Column II
A.	Endangered species	1.	Facing a very high risk of extinction in near future
B.	Threatened species	2.	Liable to become extinct
C.	Extinct species	3.	Last individual has died
D.	Vulnerable species	4.	Not critically endangered

Codes

	A	B	C	D		A	B	C	D
(a)	1	3	2	4	(b)	4	3	2	1
(c)	3	4	2	1	(d)	1	2	3	4

162. Which hormone is responsible for maintaining the circadian rhythms in non-mammalian vertebrates as well as acts as a neurosecretory transducer?

(a) Pituitary gland (b) Thymus gland
(c) Thyroid gland (d) Pineal gland

163. In which stage of childbirth uterine contractions begin marking the beginning of labour?

(a) Expulsion stage (b) Placental stage
(c) Dilation stage (d) Gestation stage

164. A farmer wished to produce bananas that were longer and sweeter. He opted for grafting method to achieve this. What are the chances of his success to achieve this target?

(a) 50% (b) 100% (c) 25% (d) Negligible

165. Ecosystem services include

(a) healthy forest ecosystem purifies air and water
(b) generate fertile soil
(c) maintain biodiversity
(d) All of the above

166. Which one of the following pairs exhibits mutualism between them?

I. Fig and *Blastophaga* wasp.
II. *Ophrys* and male *Colpa* wasp.
III. Honeybees and *Salvia*.
IV. Coral tree and squirrels.

Choose the correct option.

(a) Only I (b) I and II (c) I, II and III (d) All are correct

167. Smoking leads to which of the following diseases?

(a) Emphysema (b) Tuberculosis
(c) Bronchitis (d) Both (a) and (c)

168. The condition where the anthers do not dehisce and the germinated pollen tube pierces anther wall and enters the stigma of same flower, is known as

(a) geitonogamy (b) cleistogamy (c) herkogamy (d) dicliny

169. *Cucurbita* shows

(a) reticulate multicostate divergent type venation
(b) reticulate multicostate convergent type venation
(c) parallel unicostate venation
(d) parallel multicostate venation

170. In which type of ovule the embryo sac becomes curved-like horse shoe?
(a) Orthotropous (b) Anatropous
(c) Campylotropous (d) Amphitropous

171. Tryptophan operon in *E. coli*, is an example of
(a) positively regulated operon
(b) inducible operon
(c) repressible operon
(d) None of the above

172. Identify the correctly matched pair.
(a) Virus – Take over cellular machinery of the host cell
(b) Viroids – Modified hydrophobic glycoproteins
(c) Both (a) and (b)
(d) None of the above

173. Varun gorges upon the potato finger chips which he is very fond of. Which is the correct order of enzymes that are likely to act on it from mouth onwards?
(a) Salivary amylase – Pancreatic amylase – Disaccharidases
(b) Pancreatic amylase – Salivary amylase – Lipases
(c) Salivary maltase – Lipases – Nucleases
(d) Disaccharidases – Lipases – Nucleases

174. Identify the incorrect statement(s).
I. Bioballistic is a method in which vectors of pathogens are allowed to infect the cell to transfer the recombinant DNA.
II. Microinjection is a method when cells are bombarded with high velocity particles of gold coated with DNA.
III. Heat shock is a method when the recombinant DNA is incubated with the host cells at 80-100°C.
Choose the correct option.
(a) Only I (b) Only II (c) Only III (d) All of these

175. Genetically Modified Organisms (GMOs) are developed by *r*DNA technology and genetic engineering. They are useful in
(a) enhancing food's nutritional value
(b) reducing in the post-harvest losses
(c) increasing the tolerance power of crops against the abiotic stresses
(d) All of the above

176. Which of the following statement(s) is/are correct about tocopherols, also called 'beauty vitamins'?
(a) It is an antioxidant
(b) It protects unsaturated fatty acids and membranes
(c) Its deficiency causes sterility and muscular atrophy
(d) All of the above

177. The five kingdom system of classification is based on
(a) cell structure (b) body structure
(c) nutrition and lifestyle (d) All of these

178. The non-collapsible wall of trachea in human beings is because of the incomplete rings made of
(a) elastic cartilage (b) hyaline cartilage
(c) Both (a) and (b) (d) None of these

179. Plasmids found in prokaryotic cells, are also known as
(a) peroxisomes (b) extranuclear RNA
(c) microbodies (d) extranuclear DNA

180. Cellular totipotency was demonstrated by
(a) AV Leeuwenhoek (b) Robert Hooke
(c) FC Steward (d) Schleiden and Schwann

